

FinPilot — Study Guide (12-Hour Preparation)

Project: FinPilot – Financial Intelligence & Tax Management Platform

Stack: React • Django • PostgreSQL • MongoDB • Scikit-learn

Goal:

Learn the complete project in one day for project submission and viva.

Study Order

Project Overview



Folder Structure



Authentication



Clients



Companies



Transactions



Documents



Tax



Machine Learning



Analytics



Database



Complete Demo



Viva

Phase 1 — Project Overview (1 Hour)

Learn

- What problem does FinPilot solve?
- Why was it built?
- Who will use it?
- What are the main features?
- Why React?
- Why Django?
- Why MongoDB?
- Why PostgreSQL?
- Why Machine Learning?

Read

PROJECT_PRESENTATION.md
VIVA_PREPARATION.md
FINAL_SUBMISSION_REPORT.md

Understand

```
graph TD
    User --> Frontend
    Frontend --> Backend
    Backend --> Database
    Database --> Response
```



Frontend

Phase 2 — Folder Structure (30 Minutes)

Draw this from memory.

Frontend

```
frontend/src/  
  components/  
  context/  
  features/  
  routes/  
  lib/
```

Features

```
auth/  
clients/  
companies/  
transactions/  
documents/  
tax/  
analytics/  
ml/  
dashboard/  
profile/  
system/  
landing/  
errors/
```

Know one sentence for each folder.

Backend

```
backend/apps/  
accounts/  
clients/  
companies/  
transactions/  
documents/  
tax/  
analytics/  
ml/  
dashboard/  
system/  
mongo/
```

Know one sentence for each module.

Phase 3 — Authentication (1 Hour)

Backend Files

```
backend/apps/accounts/views.py  
backend/apps/accounts/serializers.py  
backend/apps/accounts/models.py  
backend/apps/accounts/permissions.py  
backend/apps/accounts/urls.py
```

Frontend Files

frontend/src/features/auth/

AuthContext.jsx

api.js

authorization.js

pages/

LoginPage.jsx

RegisterPage.jsx

VerifyOtpPage.jsx

ForgotPasswordPage.jsx

ResetPasswordPage.jsx

Shared

frontend/src/lib/axios.js

frontend/src/components/ProtectedRoute.jsx

frontend/src/routes/router.jsx

Learn

Registration



OTP Verification



Login



JWT



Refresh Token



Logout



Forgot Password

Phase 4 — Clients (30 Minutes)

Backend

```
backend/apps/clients/  
views.py  
services.py  
serializers.py  
urls.py
```

Frontend

```
frontend/src/features/clients/  
api.js  
hooks.js  
pages/
```

```
ClientListPage.jsx
```

```
ClientDetailPage.jsx
```

Learn

Create Client



Update Client



Delete Client



Map Client to Companies

Phase 5 — Companies (45 Minutes)

Backend

```
backend/apps/companies/
```

```
views.py
```

```
services.py
```

```
serializers.py
```

```
urls.py
```

Frontend

```
frontend/src/features/companies/
```

```
api.js
```

```
hooks.js
```

```
pages/
```

```
CompanyListPage.jsx
```

```
CompanyDetailPage.jsx
```

Learn

Create Company

↓

Update Company

↓

Delete Company

↓

Select Active Company

Phase 6 — Transactions (2 Hours)

Backend

```
backend/apps/transactions/
```

```
├──
```

```
views.py
```

```
├──
```

```
services.py
```

```
└──
```



```
serializers.py
```

```
urls.py
```

Frontend

```
frontend/src/features/transactions/
```

```
api.js
```

```
hooks.js
```

```
pages/
```

```
TransactionsPage.jsx
```

```
CategoriesPage.jsx
```

```
VendorsPage.jsx
```

```
CustomersPage.jsx
```

Learn

Transaction CRUD

↓

Categories

↓

Vendor

↓

Customer

↓

Budget

↓

CSV Import

↓

Excel Import

↓

Export

↓

Analytics Data

Ignore helper functions.

Phase 7 — Documents (45 Minutes)

Backend

```
backend/apps/documents/  
views.py  
services.py  
serializers.py
```

Frontend

```
frontend/src/features/documents/  
DocumentsPage.jsx  
DocumentDetailPage.jsx
```

Learn

Upload



Store



Duplicate Detection



Archive

Phase 8 — Tax (45 Minutes)

Backend

```
backend/apps/tax/  
views.py  
services.py  
gst_rules.py  
tds_rules.py
```

Frontend

```
frontend/src/features/tax/  
TaxCenterPage.jsx
```

Learn

GST



TDS



Income Tax



Compliance



Filing Readiness

Phase 9 — Machine Learning (45 Minutes)

Backend

```
backend/apps/ml/
```

```
train.py
```

```
predict.py
```

```
features.py
```

```
services.py
```

```
utils.py
```

Frontend

```
frontend/src/features/ml/
```

```
AIDashboardPage.jsx
```

Learn

Expense Categorization



Random Forest

Duplicate Detection



Isolation Forest

Compliance Risk



Decision Tree

Expense Forecast



Linear Regression

Tax Forecast



Linear Regression

Don't memorize algorithms.

Only know why each one is used.

Phase 10 — Analytics (45 Minutes)

Backend

```
backend/apps/analytics/
```

views.py

■

services.py

■

utils.py

Frontend

frontend/src/features/analytics/

AnalyticsDashboardPage.jsx

ExecutiveDashboardPage.jsx

ReportCenterPage.jsx

Learn

KPIs

↓

Mongo Aggregation

↓

Charts

↓

Reports

↓

Export PDF

↓

Export Excel

Phase 11 — Dashboard (30 Minutes)

Backend

```
backend/apps/dashboard/  
views.py  
services.py
```

Frontend

```
DashboardPage.jsx
```

Learn

Dashboard

↓

Statistics

↓

Recent Activity

↓

Quick Actions

Phase 12 — Routing (30 Minutes)

Read

```
frontend/src/routes/router.jsx
```

Understand

Landing



Login



Dashboard



Clients



Companies



Transactions



Tax



Analytics



ML



Settings



Logout

Phase 13 — API Flow (1 Hour)

Example

TransactionsPage.jsx

↓

hooks.js

↓

api.js

↓

Axios

↓

Django View

↓

Service

↓

MongoDB

↓

Response

↓

React Query

↓

UI

If you understand one flow,

you understand almost the whole project.

Phase 14 — Database (45 Minutes)

PostgreSQL

Know these tables

Users

EmailVerificationOTP

PasswordResetOTP

OutstandingToken

BlacklistedToken

AuditLog

Purpose

Authentication

↓

OTP

↓

JWT

↓

Audit

MongoDB

Know these collections

```
clients
client_company_mapping
company_members
companies
transactions
documents
transaction_categories
vendors
customers
parties
budgets
analytics
reports
analytics_report_log
ml_models
user_preferences
```

Purpose

Business Data

Phase 15 — Run Complete Demo (2 Hours)

Repeat this until comfortable.

```
Landing
┆
┆ ↓
┆
Register
┆
┆ ↓
┆
Verify OTP
┆
┆ ↓
┆
Login
┆
┆ ↓
┆
```



Create Client



Create Company



Add Vendor



Add Customer



Create Transaction



Import CSV



Upload Documents



Tax Center



Analytics



ML Dashboard



Train Models



Download Report



Profile



Settings



Logout

Phase 16 — Viva Preparation (1 Hour)

For every module answer:

1

What does it do?

2

Which files implement it?

3

How does data flow?

Example

Transactions

Purpose

Manage financial records.

Backend

```
views.py
services.py
```

Frontend

```
TransactionsPage.jsx
api.js
hooks.js
```

Database

```
MongoDB
transactions collection
```

Important Files

Frontend

```
routes/router.jsx
AuthContext.jsx
ProtectedRoute.jsx
DashboardLayout.jsx
ThemeContext.jsx
lib/axios.js
features/clients/api.js
```

```
features/companies/api.js
|
features/transactions/api.js
|
ClientListPage.jsx
|
CompanyListPage.jsx
|
TransactionsPage.jsx
|
DashboardPage.jsx
```

Backend

```
config/settings.py
|
config/urls.py
|
accounts/views.py
|
clients/views.py
|
companies/views.py
|
transactions/views.py
|
documents/views.py
|
tax/views.py
|
analytics/views.py
|
ml/train.py
|
ml/predict.py
```

Don't Waste Time Learning

Skip these unless asked.

Utility files

Every helper

Every regex

Every serializer field

Tailwind implementation

Every React hook

Every import statement

Small helper functions

Understand the project.

Don't memorize code.

12-Hour Timeline

Time	Task
1 Hour	Project Overview
30 min	Folder Structure
1 Hour	Authentication
30 min	Clients
45 min	Companies
2 Hours	Transactions
45 min	Documents

45 min	Tax
45 min	Machine Learning
45 min	Analytics
45 min	Database
2 Hours	Demo Practice
1 Hour	Viva Practice

Final Goal

By the end of this guide, you should be able to explain:

- Overall Architecture
- Authentication Flow
- React Frontend
- Django Backend
- PostgreSQL
- MongoDB
- Machine Learning
- Tax Module
- Analytics
- Complete User Flow

without needing to memorize every line of code.